Partial Standing Waves and Wave Reflection

Superposition of incident and reflected waves, the partial-clapotis envelope, and the reflection coefficient. Companion note to the online calculator; follows Dean & Dalrymple (1991) §4.6.

1. Incident plus reflected wave

When a progressive wave meets a structure, part of its energy is reflected. The reflected wave travels back over the incident wave, and the two superpose to form a **partial standing wave** (partial clapotis). Writing the incident wave (height H_i) travelling toward the wall and the reflected wave (height $K_r H_i$) travelling away, the surface is:

$$\eta(x,t) = (H_i/2)\cdot\cos(kx+\omega t) + (K_r H_i/2)\cdot\cos(kx-\omega t), \quad K_r = H_r/H_i \ (0 \le K_r \le 1).$$

2. The envelope: nodes and antinodes

Taking the maximum over time at each position gives the envelope of wave heights. Its value oscillates in space between a maximum at the **antinodes** and a minimum at the **nodes**:

$$H(x) = H_i\cdot\sqrt{1 + K_r^2 + 2K_r \cos 2kx}; \quad H_{max} = (1+K_r)H_i \ \text{(antinodes, } x = 0, L/2, L\ldots\text{)}, \quad H_{min} = (1-K_r)H_i \ \text{(nodes, } x = L/4, 3L/4\ldots\text{)}.$$

Antinodes recur every half wavelength and nodes lie midway between them, so the node–antinode spacing is L/4. For a perfect reflector (K_r = 1) the height doubles at the wall and vanishes at the nodes — a full standing wave (clapotis); for K_r = 0 the height is uniform (progressive wave).

3. Measuring the reflection coefficient

Because H_{max} and H_{min} depend only on K_r, the reflection coefficient is recovered from a measured envelope (the Healy 1952 method) — e.g. from a wave gauge traversed over at least a half wavelength:

$$K_r = (H_{max} - H_{min}) / (H_{max} + H_{min}). \quad \text{Reflected energy fraction} = K_r^2; \text{ transmitted or dissipated} = 1 - K_r^2.$$

4. Envelope and height distribution

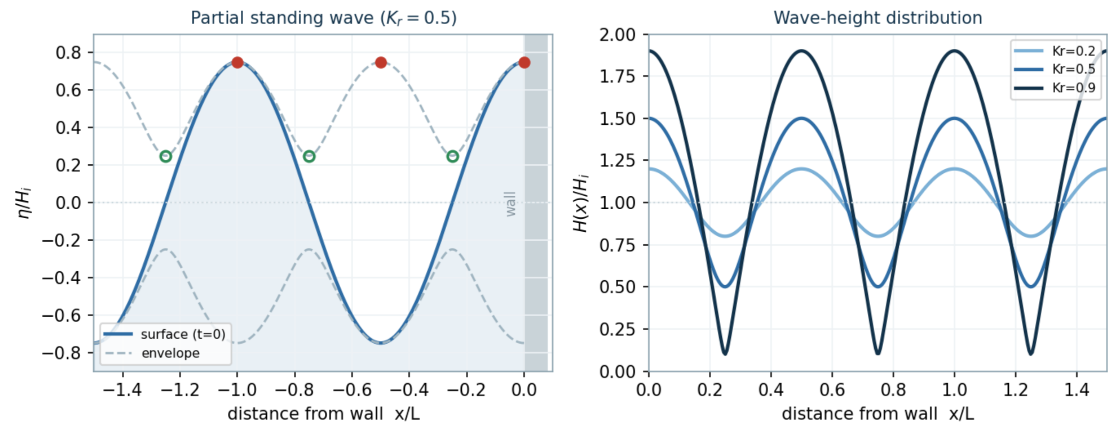

Figure 1. Left: a partial standing wave (K_r=0.5) with its envelope, antinodes (filled) and nodes (open) against a reflecting wall. Right: the wave-height distribution for several K_r — the spatial modulation deepens as reflection increases.

5. Typical reflection coefficients

Structure / shoreline	Reflection coefficient K_r
Vertical impermeable wall / caisson	0.7 – 1.0
Perforated / slotted vertical wall	0.3 – 0.6
Rubble-mound breakwater (rock armour)	0.3 – 0.6
Concrete armour-unit slope	0.25 – 0.5
Steep / reflective beach	0.2 – 0.3
Gentle / dissipative beach	0.05 – 0.15

For rock and armoured slopes K_r increases with the Iribarren number; a convenient fit is $K_r = \tanh(a\cdot\xi^b)$ (Zanuttigh & Van der Meer, 2008).

6. Worked example

Quantity	Value
Inputs	H_i = 1.5 m, T = 8 s, d = 6 m, K_r = 0.5
Wavelength, L (dispersion)	57.5 m
Reflected height, $H_r = K_r H_i$	0.75 m
Antinode height, $H_{max} = (1+K_r)H_i$	2.25 m
Node height, $H_{min} = (1–K_r)H_i$	0.75 m
Standing-wave ratio, H_{max}/H_{min}	3.0
Reflected energy, K_r^2	25 %
Antinodes / nodes from wall	0, 28.8, 57.5 m / 14.4, 43.1 m

Notation

H_i incident wave height; H_r reflected height; K_r reflection coefficient; H_{max}/H_{min} antinode/node heights; L wavelength; $k = 2\pi/L$; $\omega = 2\pi/T$; ξ Iribarren number. Linear (small-amplitude) theory; normal incidence assumed.

References

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